

MCS-013 Discrete Mathematics Assignment Solutions

Question 1

(a) What is Set? Explain use of Sets in problem solving. (3 Marks)

Definition of Set: A set is a well-defined collection of distinct objects, considered as an object in its own right. The objects in a set are called elements or members of the set. Sets are typically denoted by capital letters (A, B, C) and elements are listed within curly braces.

Example: $A = \{1, 2, 3, 4, 5\}$

Use of Sets in Problem Solving:

1. **Organization of Data:** Sets help organize and classify data into meaningful groups, making complex problems more manageable. For example, categorizing students by courses enrolled.
2. **Elimination of Duplicates:** Sets automatically handle uniqueness, ensuring each element appears only once, which is useful in database management and data cleaning.
3. **Relationship Analysis:** Sets help analyze relationships between different groups using operations like union, intersection, and difference. For example, finding common customers between two stores.
4. **Logic and Reasoning:** Sets provide a foundation for logical reasoning and can be used to solve problems involving conditions and constraints.
5. **Real-world Applications:** Sets are used in database queries, search algorithms, probability theory, and defining mathematical structures.

(b) Proper Subset and Number of Proper Subsets (2 Marks)

Proper Subset: A set A is a proper subset of set B (denoted $A \subset B$) if all elements of A are in B, but B contains at least one element not in A. In other words, $A \subseteq B$ and $A \neq B$.

Given Set: $S = \{2, 3, 4, 5, 6, 7, 8, 9\}$

Number of elements: $n = 8$

Formula for number of proper subsets: $2^n - 1$

Calculation: $2^8 - 1 = 256 - 1 = 255$ proper subsets

(We subtract 1 to exclude the set itself, as a set is not a proper subset of itself)

(c) Truth Tables (3 Marks)

i) $p \rightarrow \sim(r \wedge q) \vee (\sim p \wedge r)$

p	q	r	$r \wedge q$	$\sim(r \wedge q)$	$p \rightarrow \sim(r \wedge q)$	$\sim p$	$\sim p \wedge r$	Result
T	T	T	T	F	F	F	F	F
T	T	F	F	T	T	F	F	T
T	F	T	F	T	T	F	F	T
T	F	F	F	T	T	F	F	T
F	T	T	T	F	T	T	T	T
F	T	F	F	T	T	T	F	T
F	F	T	F	T	T	T	T	T
F	F	F	F	T	T	T	F	T

ii) $p \rightarrow (r \vee q) \vee (\sim p \vee \sim r)$

p	q	r	$r \vee q$	$p \rightarrow (r \vee q)$	$\sim p$	$\sim r$	$\sim p \vee \sim r$	Result
T	T	T	T	T	F	F	F	T
T	T	F	T	T	F	T	T	T
T	F	T	T	T	F	F	F	T
T	F	F	F	F	F	T	T	T
F	T	T	T	T	T	F	T	T
F	T	F	T	T	T	T	T	T
F	F	T	T	T	T	F	T	T
F	F	F	F	T	T	T	T	T

iii) $p \rightarrow (r \wedge \sim q)$

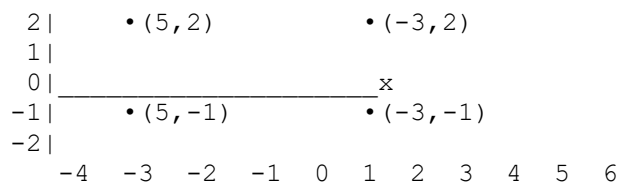
p	q	r	$\sim q$	$r \wedge \sim q$	$p \rightarrow (r \wedge \sim q)$
T	T	T	F	F	F
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	F	T

(d) Geometric Representation (2 Marks)

i) $\{5, -3\} \times \{-1, 2\}$

Cartesian Product: $\{(5, -1), (5, 2), (-3, -1), (-3, 2)\}$

Geometric Representation:

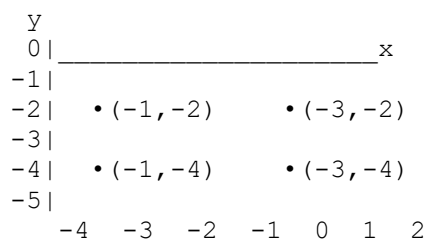


Four points plotted on the Cartesian plane.

ii) $\{-1, -3\} \times \{-2, -4\}$

Cartesian Product: $\{(-1, -2), (-1, -4), (-3, -2), (-3, -4)\}$

Geometric Representation:

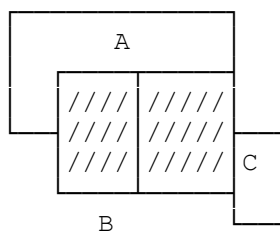


Four points in the third quadrant.

Question 2

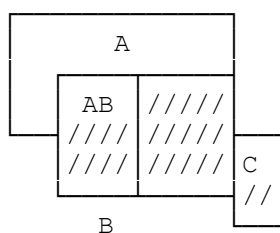
(a) Venn Diagrams (3 Marks)

i) $(A \cup B) \cap C$



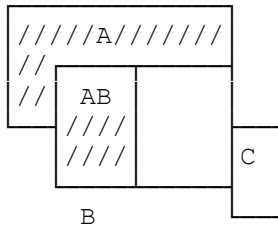
Shaded region: Areas where C overlaps with either A or B.

ii) $(A \cap B) \cup C$



Shaded region: Intersection of A and B, plus all of C.

iii) $(A \cup B) \cap C'$



Shaded region: Areas in A or B but outside C.

(b) Mathematical Statements from Symbolic Properties (2 Marks)

i) $(\forall x)(\forall y)(\forall z)P$

Statement: "For all values of x, for all values of y, and for all values of z, property P holds true."

Example: For all real numbers x, y, and z, the equation $x + (y + z) = (x + y) + z$ is true (associative property of addition).

ii) $\exists(x) \forall(y) Q$

Statement: "There exists at least one value of x such that for all values of y, property Q holds true."

Example: There exists a number $x = 0$ such that for all real numbers y, $x + y = y$.

(c) Proving $\sqrt{2}$ is Irrational (2 Marks)

Proof by Contradiction:

Assumption: Assume $\sqrt{2}$ is rational, meaning $\sqrt{2} = p/q$ where p and q are integers with no common factors (in lowest terms), and $q \neq 0$.

Steps:

1. Square both sides: $2 = p^2/q^2$
2. Multiply both sides by q^2 : $2q^2 = p^2$
3. This means p^2 is even (divisible by 2)
4. If p^2 is even, then p must be even (since $\text{odd}^2 = \text{odd}$)
5. Let $p = 2k$ for some integer k
6. Substituting: $2q^2 = (2k)^2 = 4k^2$
7. Simplifying: $q^2 = 2k^2$
8. This means q^2 is even, so q is also even

Contradiction: Both p and q are even, meaning they share a common factor of 2. This contradicts our initial assumption that p and q have no common factors.

Conclusion: Our assumption was wrong. Therefore, $\sqrt{2}$ is **irrational**.

(d) Proof by Contradiction with Example (3 Marks)

Definition: Proof by contradiction (also called indirect proof or reductio ad absurdum) is a logical method where we assume the opposite of what we want to prove, and then show that this assumption leads to a logical contradiction. This contradiction proves that our original statement must be true.

Method:

1. Assume the negation of the statement to be proved
2. Use logical deduction to derive a contradiction
3. Conclude that the assumption must be false
4. Therefore, the original statement is true

Example: Prove that there are infinitely many prime numbers

Proof:

1. **Assumption:** Assume there are finitely many primes: $p_1, p_2, p_3, \dots, p_n$
2. **Construction:** Consider the number $N = (p_1 \times p_2 \times p_3 \times \dots \times p_n) + 1$
3. **Analysis:**
 - N is larger than all known primes
 - When N is divided by any prime p_i , the remainder is 1
 - Therefore, N is not divisible by any of the known primes
4. **Two Possibilities:**
 - Either N itself is prime (contradicting that we listed all primes)
 - Or N has a prime factor not in our list (also contradicting completeness)
5. **Contradiction:** Our assumption that there are finitely many primes leads to a contradiction.
6. **Conclusion:** There must be infinitely many prime numbers.

Question 3

(a) Inclusion-Exclusion Principle with Example (2 Marks)

Inclusion-Exclusion Principle: For two sets A and B , the number of elements in their union is: $|A \cup B| = |A| + |B| - |A \cap B|$

For three sets A , B , and C : $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Example: In a class of 50 students:

- 30 students play cricket (C)
- 25 students play football (F)
- 10 students play both sports

Question: How many students play at least one sport?

Solution: $|C \cup F| = |C| + |F| - |C \cap F|$ $|C \cup F| = 30 + 25 - 10 = 45$ **students**

Therefore, 45 students play at least one sport, and 5 students play neither.

(b) Logic Circuits for Boolean Expressions (3 Marks)

i) $(x + yz) + (yz)' + (z'y)$

Simplified Expression:

- $(yz)' = y' + z'$ (De Morgan's Law)
- Expression becomes: $x + yz + y' + z' + z'y$

Logic Circuit Components:

1. AND gate for yz
2. AND gate for $z'y$
3. NOT gates for y' , z'
4. OR gate combining all terms

Description:

- Input x directly to final OR
- y AND $z \rightarrow$ to OR
- NOT $y \rightarrow$ to OR
- NOT $z \rightarrow$ to OR
- (NOT z) AND $y \rightarrow$ to OR
- Final OR gate combines all five inputs

ii) $(x'y)(xz')(y'z) + xyz$

Logic Circuit Components:

1. Three AND gates for the product terms
2. One final AND gate connecting all products
3. One AND gate for xyz
4. Final OR gate

Description:

- NOT x , then x' AND y
- x AND NOT z
- NOT y AND z
- Three outputs connected by AND

- Separately: $x \text{ AND } y \text{ AND } z$
- Both results connected by OR

(c) Tautology and Proof (3 Marks)

Definition of Tautology: A tautology is a logical statement that is always true, regardless of the truth values of its component propositions. In a truth table, a tautology has all entries in the final column as TRUE.

Statement to Prove: $(P \rightarrow Q) \vee (\sim P)$

Truth Table:

P	Q	$P \rightarrow Q$	$\sim P$	$(P \rightarrow Q) \vee (\sim P)$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Conclusion: The statement $(P \rightarrow Q) \vee (\sim P)$ is **NOT a tautology** because it is false when P is true and Q is false.

Note: The statement $P \rightarrow Q \vee \sim P$ (with different precedence) or $(\sim P) \vee Q$ would be a tautology.

(d) Symmetric Difference of Sets with Examples (2 Marks)

Definition: The symmetric difference of two sets A and B, denoted $A \Delta B$ or $A \oplus B$, is the set of elements that belong to either A or B, but not to both.

Formula: $A \Delta B = (A \cup B) - (A \cap B) = (A - B) \cup (B - A)$

Mathematical Example: If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ $A \Delta B = \{1, 2, 5, 6\}$

Real-Life Examples:

1. Social Media Friends:

- Person A's friends: {John, Mary, Tom, Lisa}
- Person B's friends: {Tom, Lisa, Sarah, Mike}
- Symmetric difference: {John, Mary, Sarah, Mike}
- Represents friends unique to each person

2. Product Features:

- Product X features: {WiFi, Bluetooth, Camera, GPS}
 - Product Y features: {Bluetooth, GPS, NFC, USB-C}
 - Symmetric difference: {WiFi, Camera, NFC, USB-C}
 - Shows features that distinguish one product from another
-

Question 4

(a) Word Formation from "EXCEPTIONAL" (2 Marks)

Word: EXCEPTIONAL **Letters:** E-X-C-E-P-T-I-O-N-A-L (11 letters)

Frequency Count:

- E: 2
- X: 1
- C: 1
- P: 1
- T: 1
- I: 1
- O: 1
- N: 1
- A: 1
- L: 1
- Total: 11 letters (with E repeated twice)

i) If each letter must be used:

Formula for permutations with repetition: $n! / (n_1! \times n_2! \times \dots \times n_k!)$

Calculation: $11! / 2! = 39,916,800 / 2 = 19,958,400$ words

ii) If some or all letters may be omitted:

For each letter, we can either:

- Use it (in any position)
- Omit it

However, E appears twice, so we need to consider:

- Use no E
- Use one E
- Use both Es

This is complex. A simpler approach:

Total possible selections:

- For letters appearing once: 2 choices each (use or don't use) = 2^9
- For E (appears twice): 3 choices (0 Es, 1 E, or 2 Es)

Total subsets: $2^9 \times 3 = 512 \times 3 = 1,536$ possible selections

For each non-empty selection, we need to arrange the letters:

- This requires summing permutations for each possible combination
- Exact calculation: Σ (arrangements for each valid subset)

Approximate answer: For each of the 1,535 non-empty subsets (excluding empty set), calculate permutations considering repetitions.

Simplified: Since exact calculation is complex, the answer involves calculating permutations for each possible subset length from 1 to 11, considering E's repetition.

(b) Relations and Types (8 Marks)

Definition of Relation: A relation R from set A to set B is a subset of the Cartesian product $A \times B$. It defines a relationship between elements of A and elements of B . In other words, if $(a, b) \in R$, we say "a is related to b" and write aRb .

Example: Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$ $A \times B = \{(1,x), (1,y), (2,x), (2,y), (3,x), (3,y)\}$ A relation R could be: $R = \{(1,x), (2,y), (3,x)\}$

Types of Relations:

1. Empty Relation

Definition: A relation R on set A is empty if no element of A is related to any element of A .

Example: $A = \{1, 2, 3\}$, $R = \{\}$ (no pairs)

2. Universal Relation

Definition: A relation R on set A is universal if every element of A is related to every element of A . **Example:** $A = \{1, 2\}$, $R = \{(1,1), (1,2), (2,1), (2,2)\} = A \times A$

3. Reflexive Relation

Definition: A relation R on set A is reflexive if every element is related to itself: $(a, a) \in R$ for all $a \in A$. **Example:** $A = \{1, 2, 3\}$, $R = \{(1,1), (2,2), (3,3), (1,2)\}$ Every element has a pair with itself.

4. Symmetric Relation

Definition: A relation R on set A is symmetric if whenever $(a, b) \in R$, then $(b, a) \in R$.

Example: $A = \{1, 2, 3\}$, $R = \{(1,2), (2,1), (2,3), (3,2)\}$ "is a sibling of" is a symmetric relation.

5. Antisymmetric Relation

Definition: A relation R on set A is antisymmetric if whenever $(a, b) \in R$ and $(b, a) \in R$, then $a = b$. **Example:** $A = \{1, 2, 3\}$, $R = \{(1,1), (1,2), (2,3)\}$ "is less than or equal to" ($\leq$) is antisymmetric.

6. Transitive Relation

Definition: A relation R on set A is transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$. **Example:** $A = \{1, 2, 3, 4\}$, $R = \{(1,2), (2,3), (1,3), (2,4), (1,4)\}$ "is an ancestor of" is a transitive relation.

7. Equivalence Relation

Definition: A relation that is reflexive, symmetric, and transitive. **Example:** $A = \{1, 2, 3, 4\}$ $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)\}$ "has the same parity as" is an equivalence relation.

8. Partial Order Relation

Definition: A relation that is reflexive, antisymmetric, and transitive. **Example:** On set $A = \{1, 2, 3, 4, 6, 12\}$ $R = \text{"divides" relation: } \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,12), (2,2), (2,4), (2,6), (2,12), (3,3), (3,6), (3,12), (4,4), (4,12), (6,6), (6,12), (12,12)\}$

Question 5

(a) Professional Committee Formation (3 Marks)

Given:

- 8 Managers available
- 6 Public Servants available
- 8 IT Professionals available
- Committee needs: 8 people total with at least 4 Managers, at least 2 Public Servants, and 2 IT Professionals

Valid Combinations: Since we need at least $4M + 2P + 2IT = 8$ people minimum, and we need exactly 8:

Managers	Public Servants	IT Professionals	Total
4	2	2	8
4	3	1	8
5	2	1	8
4	4	0	8

Wait - we need "2 IT Professionals" which means at least 2, so:

Valid combinations:

1. 4M, 2P, 2IT: $C(8,4) \times C(6,2) \times C(8,2)$
2. 5M, 2P, 1IT: Not valid (only 1 IT)
3. 6M, 2P, 0IT: Not valid (no IT)

Actually, re-reading: "at least 2 IT Professionals" means ≥ 2 .

Valid Combinations:

1. **(4M, 2P, 2IT):** $C(8,4) \times C(6,2) \times C(8,2) = 70 \times 15 \times 28 = 29,400$
2. **(4M, 3P, 1IT):** Not valid - needs at least 2 IT
3. **(5M, 2P, 1IT):** Not valid - needs at least 2 IT
4. **(4M, 4P, 0IT):** Not valid - needs at least 2 IT

Correction - let's properly identify all valid combinations for exactly 8 people:

- 4M, 2P, 2IT ✓
- 5M, 2P, 1IT ✗ (only 1 IT)
- 6M, 2P, 0IT ✗ (no IT)

Since we must have at least 2 of each critical category and total 8: Only **(4M, 2P, 2IT)** satisfies all constraints.

Answer: $C(8,4) \times C(6,2) \times C(8,2) = 70 \times 15 \times 28 = \mathbf{29,400}$ committees

(b) Probability with Mutually Exclusive Events (2 Marks)

Given:

- $P(A) = 1/3$
- $P(B) = 1/4$
- A and B are mutually exclusive

For Mutually Exclusive Events:

- $P(A \cap B) = 0$ (they cannot occur together)
- $P(A \cup B) = P(A) + P(B)$

Calculations:

$$P(A \cup B) = 1/3 + 1/4 = 4/12 + 3/12 = \mathbf{7/12}$$

$$P(A \cap B) = \mathbf{0}$$

Why? Because A and B are mutually exclusive events, meaning they cannot occur simultaneously. By definition, the probability of both occurring together is zero.

(c) Pascal's Triangle (2 Marks)

Definition: Pascal's Triangle is a triangular array of binomial coefficients. Each number is the sum of the two numbers directly above it.

Structure:

$$\begin{array}{ccccc} & & 1 & & \\ & 1 & & 1 & \\ 1 & & 2 & & 1 \end{array}$$

		1		3		3		1		
	1		4		6		4		1	
1		5		10		10		5		1
1	6	15	20	15	6	1				

Properties:

1. Each row represents coefficients of $(a + b)^n$ expansion
2. Entry at row n , position k is $C(n,k)$ or nC_k
3. Each number = sum of two numbers above it
4. Row sums equal 2^n
5. Symmetric about the vertical center

Applications:

- Binomial expansion: $(x + y)^4 = 1x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + 1y^4$
- Combinatorics: $C(n,k)$ values
- Probability theory
- Number theory patterns

(d) Finding Inverse of a Function with Example (3 Marks)

Definition: The inverse of a function f , denoted f^{-1} , is a function that "reverses" f . If $f(a) = b$, then $f^{-1}(b) = a$.

Conditions for Inverse to Exist:

- Function must be one-to-one (injective)
- Function must be onto (surjective)
- Function must be bijective

Method to Find Inverse:

1. Replace $f(x)$ with y
2. Swap x and y
3. Solve for y
4. Replace y with $f^{-1}(x)$

Example:

Find the inverse of $f(x) = 2x + 3$

Step 1: Write as $y = 2x + 3$

Step 2: Swap x and y : $x = 2y + 3$

Step 3: Solve for y :

- $x = 2y + 3$
- $x - 3 = 2y$

- $y = (x - 3)/2$

Step 4: Write inverse: $f^{-1}(x) = (x - 3)/2$

Verification:

- $f(f^{-1}(x)) = f((x-3)/2) = 2((x-3)/2) + 3 = x - 3 + 3 = x \checkmark$
- $f^{-1}(f(x)) = f^{-1}(2x+3) = ((2x+3)-3)/2 = 2x/2 = x \checkmark$

Another Example with Restricted Domain:

$f(x) = x^2, x \geq 0$

Step 1: $y = x^2$

Step 2: $x = y^2$

Step 3: $y = \sqrt{x}$ (taking positive root since $x \geq 0$)

Step 4: $f^{-1}(x) = \sqrt{x}, x \geq 0$

Question 6

(a) Distributing Items into Boxes (2 Marks)

Given: 25 distinct items, 7 distinct boxes

i) At least three empty boxes

Method: Use inclusion-exclusion or complementary counting

If at least 3 boxes are empty, we use at most 4 boxes.

Cases:

- Exactly 4 boxes used
- Exactly 3 boxes used
- Exactly 2 boxes used
- Exactly 1 box used

Using Stirling numbers and combinations:

Number of ways to use exactly k boxes:

- Choose k boxes: $C(7, k)$
- Distribute 25 items into k non-empty boxes: $S(25, k) \times k!$ (where $S(n, k)$ is Stirling number of second kind)

For exactly k boxes:

- $k=1: C(7,1) \times 1! = 7$
- $k=2: C(7,2) \times S(25,2) \times 2!$
- $k=3: C(7,3) \times S(25,3) \times 3!$
- $k=4: C(7,4) \times S(25,4) \times 4!$

Answer: This requires Stirling numbers calculation, which is complex.

Alternative formula: At most 4 boxes used = $\sum_{k=1}^4 C(7,k) \times k! \times S(25,k)$

ii) No empty box

All 7 boxes must contain at least one item.

Formula: Use Stirling number of second kind $S(25,7) \times 7!$ where $S(25,7)$ distributes items into non-empty groups

Or using inclusion-exclusion: $7^{25} - C(7,1) \times 6^{25} + C(7,2) \times 5^{25} - C(7,3) \times 4^{25} + C(7,4) \times 3^{25} - C(7,5) \times 2^{25} + C(7,6) \times 1^{25}$

Simplified answer: $7! \times S(25,7)$ or using surjective function formula

(b) Properties of Set (2 Marks)

Fundamental Properties:

1. **Uniqueness:** Each element appears only once in a set
2. **Unordered:** Order of elements doesn't matter $\{1,2,3\} = \{3,1,2\}$
3. **Commutative Property:**
 - $A \cup B = B \cup A$
 - $A \cap B = B \cap A$
4. **Associative Property:**
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - $(A \cap B) \cap C = A \cap (B \cap C)$
5. **Distributive Property:**
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
6. **Identity Property:**
 - $A \cup \emptyset = A$ (empty set is identity for union)
 - $A \cap U = A$ (universal set is identity for intersection)
7. **Complement Property:**
 - $A \cup A' = U$ (universal set)
 - $A \cap A' = \emptyset$ (empty set)